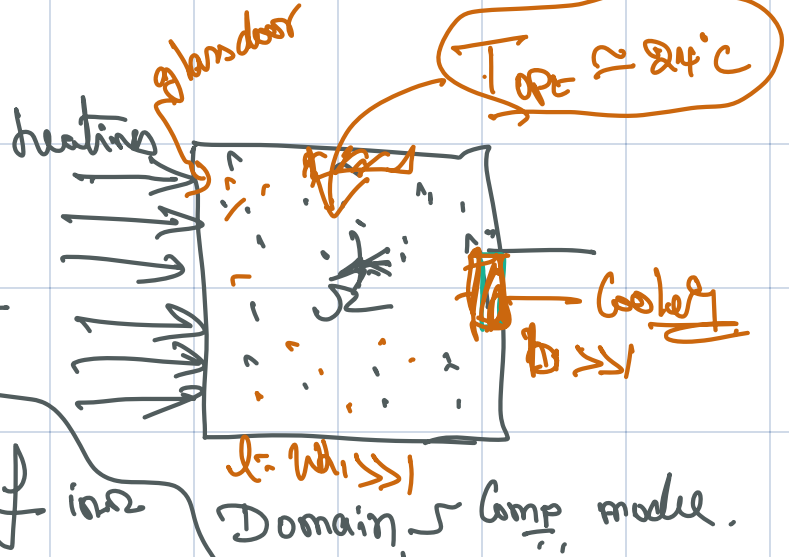


Modeling & Simulation



$$\frac{\partial T}{\partial t} - \epsilon \nabla^2 T + b \cdot \nabla T = f \text{ inside}$$

$$u = u_D \text{ on } \partial \Omega$$

Cooling
here

Chip
Casing
Battery

T - Temperature, $\epsilon \rightarrow$ Thermal Conductivity

b - Advection, f = Source

Sol 1: Analytical Sol. (not possible)

Sol 2: Numerical methods (FD, FE, FM)
(expensive)

Sol 3: Can AI be used to find sol for ①

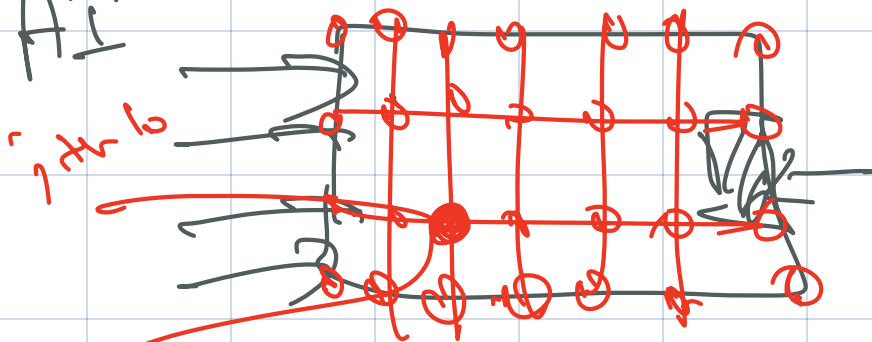
Let $b=0$, & $\varepsilon=1$, $\frac{\partial T}{\partial \tau}=0$

Temp is indep of time
 i.e. Steady State

$$-\Delta T = \frac{1}{k} \text{ in } \Omega$$

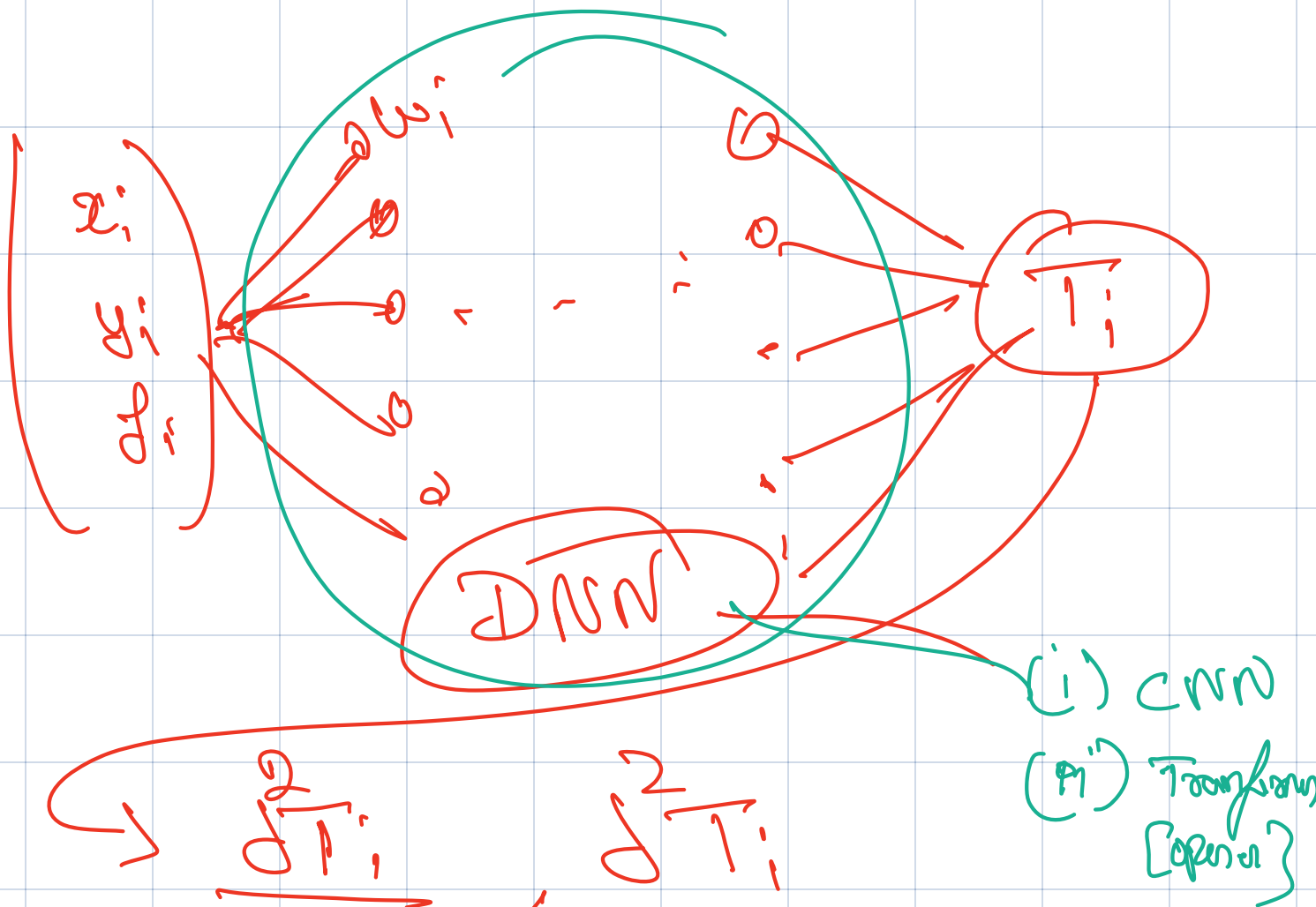
$u = u_D$

AI



(x_i, y_i, f_i)

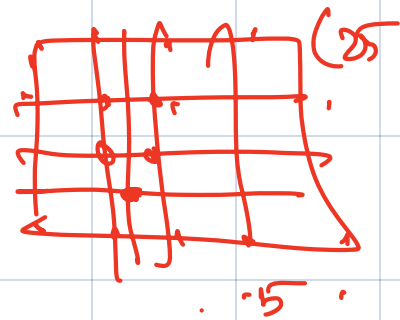
$i=1$ to 1000



$$\min L(\text{error}) = \frac{1}{1000} \sum_{i=1}^{1000} \|f_i + \Delta T_i\| \approx 0.$$

$$-\Delta T = \int n \, d\Omega$$

$$u = u_0 \sin \omega t$$



Classical N.M

$$AT = b$$

System of Algebraic eq.

Solver $\leftarrow$ expensive/time
Gross

$A_{25 \times 25}$

Again solve

NN/AI

Pre-training AI model

Inference.

10x to 400x
faster inference
than solving
 $Ax=b$ each time!

PI NNs
specific appl:
or charac.

FNO
Givov
+
arXiv
Growth.